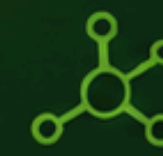




Fugacity- ML Hackathon 2026

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Problem Statement: Computational Latency Barrier



Full CFD reactor simulations are too computationally slow for real-time control, creating a bottleneck that forces a trade-off between physical accuracy and operational speed.



This is a textbook **series reaction** network: a reactant is converted into an intermediate that further reacts to form another product.

TARGET OUTPUT: OVERALL YIELD (%) OF THE DESIRED PRODUCT **B**

THE CORE PROBLEM

- In fine chemical and pharmaceutical manufacturing, lost selectivity is a critical process-safety hazard. Under thermal upset conditions, a drop in selectivity often triggers runaway secondary reactions, **where the target product degrades via competing exothermic pathways.**
- A central challenge in industrial reaction engineering is navigating the intrinsic trade-off between conversion and selectivity, where maximizing reactant conversion often accelerates the over-reaction and degradation of the target product. The kinetic modeling framework presented here explicitly accounts for these competing reaction pathways to enable optimal, real-time operating conditions.
- Full CFD and kinetic simulations are too slow for real-time control, forcing plants to operate conservatively below peak efficiency.

INDUSTRY RELEVANCE

1 THE INDUSTRIAL STANDARD
Modern manufacturing relies on **soft sensors**, software-driven models that transform accessible operational metrics into real-time estimates of hard-to-measure process outputs.

THE HYBRID PARADIGM
Industry best practice is rapidly shifting toward hybrid physics-ML soft sensors, which embed first-principles domain knowledge (conservation laws, kinetics) directly into data-driven architectures to ensure stable, bounded predictions at control-loop speeds.

2

3 MARKET SCALE
Driven by pharmaceutical adoption of continuous processing for superior yield control, the global flow chemistry market is expanding from **\$1.76 billion (2023)** to **\$3.75 billion by 2030.**

REAL INDUSTRIAL ANALOGS OF THIS PATTERN

Ethylene → ethylene oxide (desired) → over-oxidized to CO₂/H₂O (undesired)

Toluene → benzaldehyde (desired) → over-oxidized to benzoic acid (undesired)

A fast, physics-informed surrogate model replaces static margins with dynamic real-time optimization, i.e delivering maximum yield while preserving thermal stability.

Underlying Chemistry: First-order, non-Isothermal tubular reactor



Temperature-dependent reaction kinetics, residence time, and reactor conditions govern the conversion of reactants through intermediate species to the desired product.

GOVERNING PHYSICS

- Kinetics:** Closed-form solution for consecutive first-order reactions:
 $\text{yield}(\tau, T) = 100 \cdot [k_1 / (k_2 - k_1)] \cdot [e^{(-k_1 \cdot \tau)} - e^{(-k_2 \cdot \tau)}]$
where $k = A \cdot e^{(-B/T)}$
- Heat Transfer:** The feed doesn't react at a fixed temperature, it heats from **inlet temperature** toward **jacket temperature** along the tube; modelled with an NTU-style exponential approach to get an **effective reaction temperature**.

KEY ASSUMPTIONS

- First order Kinetics, both steps:** rate $\propto$ concentration
- Plug Flow:** Every fluid element moves at the same velocity without mixing, so all molecules experience the same residence time.
- NTU-style temperature relaxation:** Fluid temperature relaxes exponentially from inlet toward jacket temperature along the length, not a fixed average.
- Constant cross-section:** Area is not mentioned and is hence assumed constant. It silently gets absorbed in the pre-exponential factors A1 & A2.

IMPORTANT VARIABLES

Effective Temperature profile ($T_{\text{eff}}[\tau]$)
inlet_temperature_K & jacket_temperature_K drive the reactor's internal temperature via heat transfer

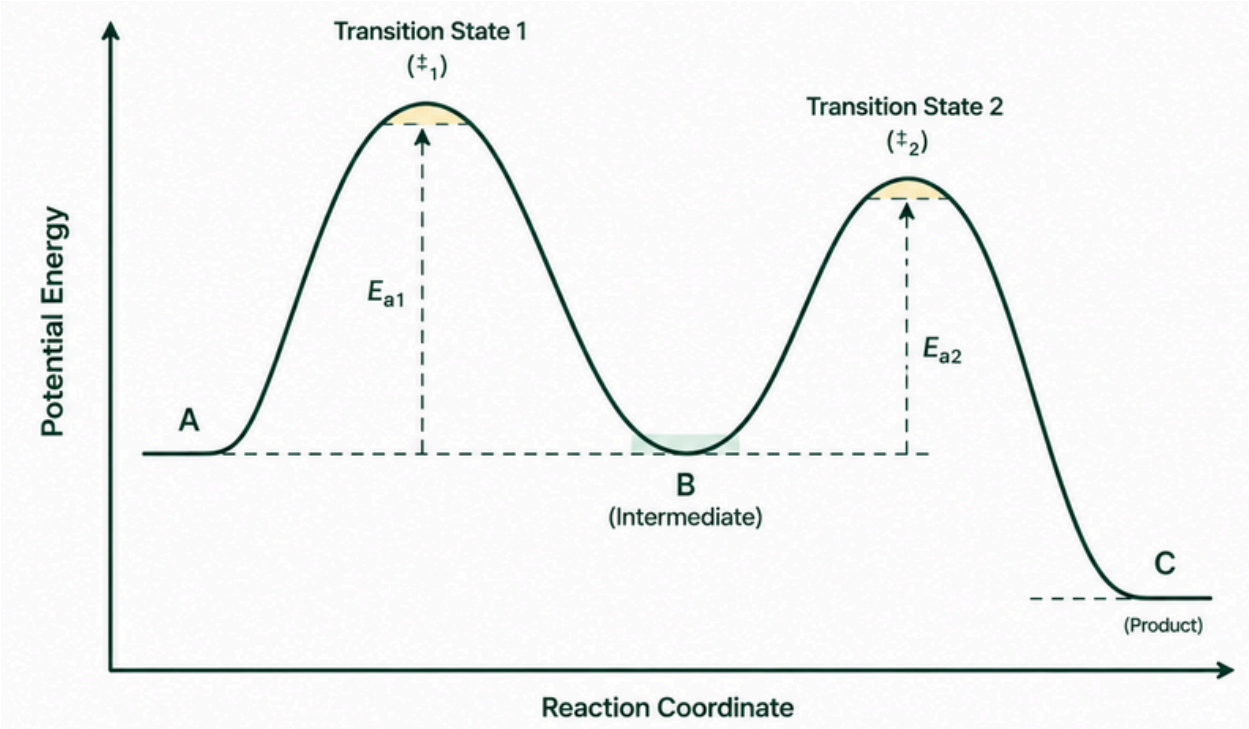
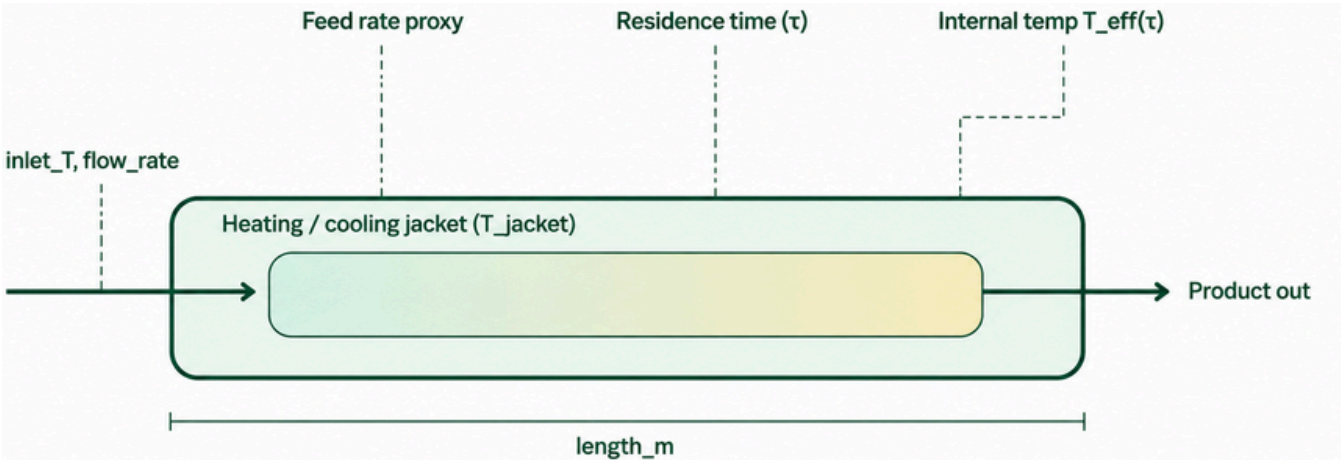
$T_{\text{eff}}(\tau) = T_{\text{jacket}} - (T_{\text{jacket}} - T_{\text{inlet}}) \cdot e^{(-c \cdot \tau)}$

$\tau = \frac{\text{length}}{\text{flow_rate}}$

Residence Time (τ)
length_m & flow_rate_L_min collectively define how long the reaction has to run in the reactor

Molar Throughput Proxy
flow_rate_L_min & concentration_mol_L is used as a molar throughput proxy (doesn't affect yield %)

$\text{molar_throughput} = \text{flow_rate} \cdot \text{concentration}$



Feature	Physically Controls	Formula
residence_time (τ)	Contact time available for reaction	length / flow_rate
delta_T	Strength & direction of the heating/cooling driving force	jacket_T - inlet_T
molar_throughput	Absolute feed rate of A (production scale, not yield %)	flow_rate $\times$ concentration
temp_ratio	Scale-free heating indicator	jacket_T / inlet_T
residence_time_sq	Lets a flexible model approximate the curvature of $\exp(-k\tau)$, which is nonlinear in τ	τ^2
heating_rate_proxy	Rate of temperature change the fluid experiences	delta_T / residence_time

Methodology: A physics-informed hybrid

A mechanistic kinetics model grounded in the reactor's real reaction chemistry generates a first-pass yield prediction, which a small neural network then refines.

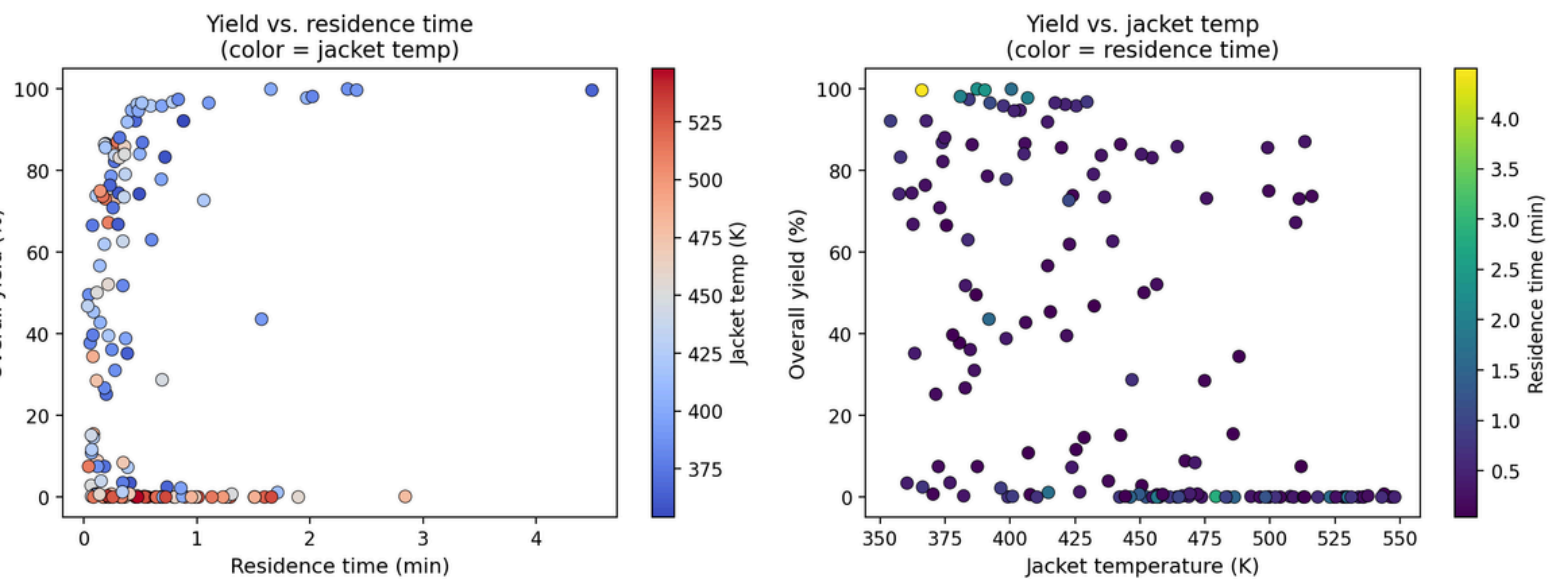
EXPLORATORY ANALYSIS

overall yield is heavily non-normal

25th pct: **0.01%**

median: **15.3%**

75th pct: **74.9%**



The rise-then-fall hump denotes a consecutive $A \rightarrow B \rightarrow C$ reaction, and the direct empirical basis for choosing that mechanism

FEATURE ENGINEERING

20 total inputs to the ML layer

9 physics derived features: residence_time, delta_T, molar_throughput, temp_ratio, 3 log-transforms, residence_time², heating_rate_proxy

6 more features derived from the mechanistic sub-model itself: T_{eff}, 1/T_{eff}, 1/T_{eff}², distance from optimal residence time, theoretical yield ceiling, and the mechanistic model's own prediction

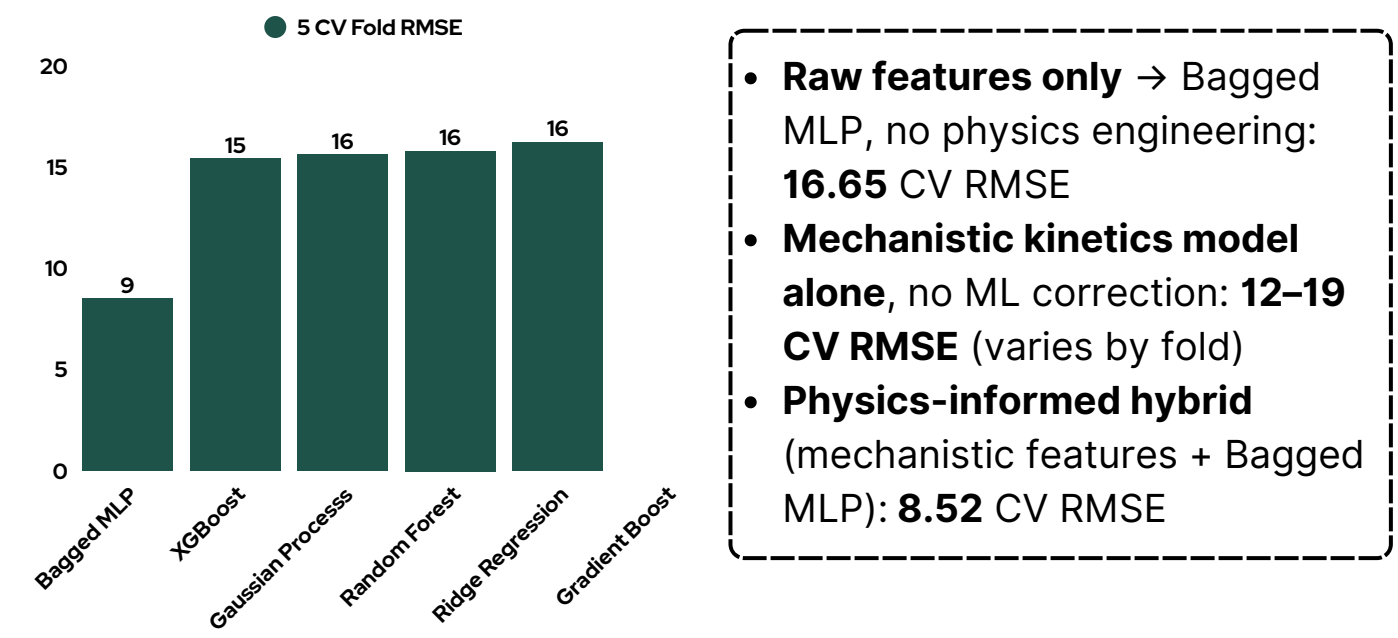
MODELS CONSIDERED

Tested 5 alternative regressors, **Ridge**, **Random Forest**, **Gradient Boosting**, **XGBoost**, **Gaussian Process** against the identical physics-engineered feature set and identical CV folds used for the final model

FINAL MODEL

Mechanistic kinetics + NTU heat-transfer sub-model feeding a Bagged MLP (2 hidden layers, 16 and 8 units, ReLU activation) with L2 regularization (alpha=2.0), lbfgs solver and predictions averaged.

BASELINE COMPARISON



VALIDATION METHODOLOGY

5-fold CV, mechanistic model refit independently inside each fold's training split only (no leakage into validation folds)

Why Not Tree-Based Models?

Once the non-isothermal temperature correction is applied, the underlying yield surface as a function of the features is smooth. Tree-based models approximate smooth functions with **axis-aligned step splits**. With only 150 samples, they lack the data needed to approximate smooth curves accurately without **overfitting**.

Why Bagged Multi Level Perceptron?

A small Multi-Layer Perceptron with smooth ReLU activations naturally **captures continuous physical relationships** better than decision trees on small datasets. Using the **L-BFGS solver for reliable convergence**, the model ensembles **15 independently seeded networks** to apply bagging principles, eliminating weight-initialization variance and stabilizing final predictions.

Model Performance: Validation & Results



Final model: 8.52 CV RMSE (physics informed hybrid) vs 16.65 CV RMSE baseline (raw features, no physics); a 49% error reduction from embedding reaction kinetics with features.

LEADERBOARD PERFORMANCE

Rank: **55/374**

RMSE: **14.8739**

MAE: **6.7164**

R²: **0.8298**

TOP PREDICTIVE FEATURES

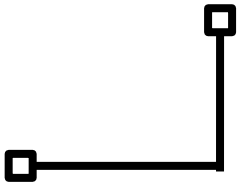
Feature	Importance
kin_pred (mechanistic model output)	30.64
log_residence_time	15.57
inlet_temperature_K	12.37
delta_T	11.77
temp_ratio	10.57

The mechanistic prediction alone carries ~2x the weight of the next-best feature; the physics is doing most of the work, and the MLP is mainly correcting it

VALIDATION RESULTS

MEAN CV RMSE: **8.519**, STD: **2.659**

Fold 1	Fold 2
8.908	7.032
Fold 3	Fold 4
8.122	5.284
Fold 5	
13.248	



Fold 5 (13.25) is much worse than fold 4 (5.28); with only 150 rows, single folds can land on hard subregions (near the yield collapse past T★).

ERROR ANALYSIS & PHYSICAL CONSISTENCY

CV (8.52) vs. leaderboard (14.87)

With n=150 training rows, a 5-fold CV estimate has real variance, and the hidden 50-row test can sit disproportionality in the steep-collapse region near T★ (506.6 K) where small input errors cause large yield swings.

Sanity- sweep check: Consistency evidence

Sweeping jacket temperature while holding other inputs at median gives a predicted yield curve that rises to a peak (~96% around 409 K), then collapses sharply past ~450 K, bottoming out near the fitted crossover temperature T★ = 506.6 K

Parameter	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Range
Ea1 (kJ/mol)	53.1	54.2	55.6	56.4	54.8	53.1 - 56.4
Ea2 (kJ/mol)	104.1	104.0	104.3	104.5	104.2	104.0 - 104.5
T* (K)	500	506	508	513	506	500 - 513

Conclusion : Multi-start kinetic fitting is not the dominant source of instability

ARRHENIUS COMPENSATION

Widening the bound to 60 significantly reduces training SSE and reveals an interior optimum rather than a boundary solution. However, **logA₂ and B₂ shift substantially together**, indicating **Arrhenius compensation**- multiple parameter pairs can produce nearly identical k(T) profiles over a limited temperature range.

Failure & Application: Where The Model Breaks



A physics-informed model is not fail-proof, knowing where it breaks, and why, is part of the engineering.

CHEMICAL
PROCESS DRIFT

OBSERVED
PREDICTION ERROR

ML / DATA
LIMITATION

- **Uniform degradation everywhere:** ML/data issue
- **Degradation concentrated in the 440–480K collapse band:** Could reflect either a sharp physics transition or a breakdown of the assumed chemistry.

CHEMICAL/PROCESS RISKS

- **Feed or catalyst drift:** silently shifts the true E_{a1} , E_{a2} away from what we fitted, biasing every prediction without any input feature signaling it.
- **Fouling or scaling:** changes the real heat-transfer rate at the jacket wall, invalidating our fitted NTU heating constant.
- **Non-ideal flow or reaction order $\neq 1$:** breaks the core plug-flow, first-order kinetics assumption the whole model rests on.

REAL-WORLD DEPLOYMENT NOTE

Suited as a **real-time yield surrogate** for plant optimization — not a replacement for periodic re-calibration against ground-truth samples.

ML/ DATA RISKS

- **Insufficient training data generally:** 150 rows for a 20-feature hybrid model is thin; permutation importance and fold performance would likely stabilize with more data.
- **Sharp yield-collapse zone:** near the threshold, a small input error produces a large output swing.
- **Bagging $\neq$ robustness to drift:** averaging seeds reduces initialization variance, not genuine distribution shift.

WHEN TO RE-VALIDATE

Whenever feed composition, catalyst batch, or the operating temperature range shifts materially from training data.

NUMERICAL SENSIVITY LIMITATION

Across different environments, the same pipeline produced slightly different predictions despite consistent overall trends. This was attributed to numerical differences arising from NumPy, SciPy, scikit-learn, and BLAS implementations

Root Cause Analysis:

The source of the reproducibility issue was isolated to the **neural-network stage of the pipeline**.

Even when near-identical mechanistic parameters were supplied to the MLP, the resulting predictions could differ substantially. More importantly, introducing an extremely small relative perturbation of only **(10^{-12})** into the MLP inputs resulted huge changes.

Why the MLP behaves this way

The **(16,8) MLP** has **~481** trainable parameters for only 150 observations, achieving a very low training RMSE. This near-interpolation, combined with **2000 LBFGS iterations** and **no early stopping**, makes predictions highly sensitive to tiny numerical differences during optimization. As a result, different environments can converge to different weight configurations with similar training performance but different test predictions.

Impact on Reproducibility

The model's validation metrics and mechanistic parameters are substantially more stable than its exact test predictions. The cross-validation RMSE and the fitted mechanistic parameters were reproducible, whereas the **exact leaderboard prediction values were sensitive to the numerical environment**. Exact reproduction of the submitted predictions cannot be guaranteed across different numerical environments.

An aerial view of a sustainable energy farm. The landscape is covered with rows of solar panels. A large white wind turbine stands on the left. In the center-right, there are several green cylindrical storage tanks for hydrogen, with labels like 'H2' and '0 330 023'. The entire scene is overlaid with a digital network interface, featuring glowing blue lines, nodes, and icons representing energy flow and data. A bar chart icon is visible in the top right corner. The word 'THANK YOU' is prominently displayed in the center in a large, white, sans-serif font.

THANK YOU